

Hidden Nesting, the Fluctuation Probe & Indeterminate Forms

Convergent resistance series, quantized erasure, and the cumulant ladder as a probe into hidden depth

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Status. Draft complete — sections 1–8 with a verbatim appendix. Two Leads resolved: both parked (§6 the cumulant ladder, §7 the primordial indeterminate). Nothing here is promoted to the project spine; findings remain in this companion note unless later judged load-bearing to the core narrative.

1. Scope and boundaries

What this note does. This note follows a single thread through four rooms.

- *Hidden nesting.* An infinite cascade of conditional erasures can carry exactly the same expected heat as a single hard erasure, so an ensemble mean can conceal unbounded machinery.
- *Quantization.* A hard erasure event comes in integer multiples of the genesis bit $\ln 2$, so a single-run heat reading bounds the number of erasure events — *actual* erasures cannot hide.
- *The fluctuation probe.* Where the mean is blind, the variance is not, and each higher cumulant is a further handle on the hidden depth — a ladder.
- *Indeterminate forms.* The classical indeterminate forms of analysis are the same shadow-underdetermination surfacing inside the calculus — a value that fails to fix its process.

The organizing claim: **convergent resistance series are the barrier’s hiding mechanism**, and the probe’s route into what they hide is the cumulant ladder. [Reading]

Regime 1 and Regime 2 — this note is the distinction at work. The project separates two stances. *Regime 1* runs forward from genesis: a quantized unfolding, structure arriving one genesis bit at a time, events discrete and integer. *Regime 2* works backward from probe interception: realized states assumed, ensembles formed, averages taken — and the continuum appears only here, as a construct of averaging. This note stages the two directly against each other: the hiding experiment’s cascade is a stream of discrete, integer-cost events (the Regime-1 face), while every heat *reading* — mean, variance, higher cumulants — is a Regime-2 ensemble quantity. The banked one-liner, following the framework: *the continuum is probe-manufactured; the system’s own ledger is integer multiples of the genesis bit*. Everything measured in this note is Regime 2; the “machines” are shadow-facing models of erasure processes, not a view behind the barrier.

Exact identities are not the content. Several results here are exact arithmetic facts — $\ln 2 = \sum_{n=1}^{\infty} 1/(n \cdot 2^n)$, the cascade variance $(2/3) \cdot \ln^2 2$, the eight indeterminate forms. These are true by computation, not discoveries; as in the companion note, where a ledger closes the closure is the entropy chain rule and is guaranteed. The empirical content is always what these quantities

reveal — that a mean cannot separate one erasure from an infinite cascade, that a variance can, that path-dependence picks out exactly the classical indeterminates. The identities are the instruments, not the findings.

Boundaries.

1. **No open-problem leverage.** Nothing here bears on any open problem — no Riemann Hypothesis, Goldbach, or P-versus-NP relevance, and no proof-discovery mechanism. The convergent series that emblematises the hiding mechanism — $\Omega = \sum 2^{-|p|}$, finite-looking magnitude over unbounded depth — is cited, not leveraged.
2. **The machines are illustrative.** The two-machine constructions are toy models chosen to make a conceptual point exactly computable; they are not physical claims, and no thermodynamic measurement is asserted.
3. **Precision convention.** A reported 0.000e+00 means agreement to within double-precision arithmetic, not a proof of exact equality; genuinely exact identities are flagged as such and checked symbolically or in exact rationals where it matters.
4. **Cartography, not evidence (as in the companion note).** Everything is Regime-2 and shadow-side; the pole arithmetic of $\{0, 1, \infty\}$ and the erasure machines say nothing about whether the genesis system holds beyond the axioms that posit it.

Grading. Every claim carries an inline grade — [Fact], [Reading], [Lead] — as throughout the project. This note carries two Leads: the *cumulant ladder* (whether matching d cumulants is what depth- d hiding costs) and the *primordial indeterminate* (whether the pre-distinction state admits a formal residue in the pole-arithmetic frame). Both are resolved — promoted, killed, or explicitly parked — before the note closes.

2. Indeterminate forms as the discontinuity set of pole arithmetic

The setup. Take three distinguished points — the poles $\{0, 1, \infty\}$ — and the four operations $\{-, \times, \div, \wedge\}$. The poles are the framework’s own distinguished points: the lattice bottom and top, 0 and ∞ reciprocal images of one another, 1 the unit between them; as Möbius points $\{0, 1, \infty\}$ is the orbit of the anharmonic (S) group generated by $z \mapsto 1/z$ and $z \mapsto 1-z$. **[Reading** for the framework identification; the S orbit is classical.**]**

The test. For each ordered triple (pole, op, pole), approach the pole values along several path families — polynomial rates $(t, t^2, \dots)$ and exponential rates $(e^{-1/t}, e^{1/t})$ — and take exact symbolic limits. A combination is **path-dependent** exactly when distinct paths give distinct limits: the endpoint value fails to determine the process. The exponential families are essential — without them 0^∞ and ∞^0 each collapse to the single limit 1 and would read as determinate. Everything here is Regime-2 and shadow-side: limits of realised paths, nothing at the genesis layer.

Result. [Fact] The path-dependent set is exactly the eight classical indeterminate forms — $\infty-\infty$, $0\cdot\infty$, $\infty\cdot 0$, $0/0$, ∞/∞ , 0^∞ , 1^∞ , ∞^0 — and nothing else (Exact match: True). Every one of the other 28 combinations returns a single value along every path. Path-dependence, defined with no prior notion of “indeterminate,” reproduces the classical list on the nose.

Two archetypes (by hand). [Fact] Treating multiplication as commutative leaves seven distinct forms, and these reduce to two. Division by a pole is multiplication by its reciprocal, so $0/0 = 0\cdot\infty$ and $\infty/\infty = \infty\cdot 0$; and each exponential form has an exponent that is itself a $0\cdot\infty$ product, since a^b

$= e^{\{b \cdot \ln a\}}$ gives exponents $0 \cdot \ln 0$ (for $0 \sim 0$), $\infty \cdot \ln 1$ (for $1 \sim \infty$), and $0 \cdot \ln \infty$ (for $\infty \sim 0$). Every multiplicative and exponential indeterminate therefore reduces to the single archetype $0 \cdot \infty$, leaving $\infty - \infty$ as the one irreducibly additive archetype. Two archetypes, not eight.

Under the resistance map. [Fact, by hand $\rightarrow$ Reading] Send each value to its resistance, $r = -\ln v$. The three poles become $0 \rightarrow +\infty$, $1 \rightarrow 0$, $\infty \rightarrow -\infty$, and multiplication of values becomes addition of resistances. The multiplicative archetype $0 \cdot \infty$ then reads, in resistance coordinates, as $(+\infty) + (-\infty) = \infty - \infty$ — the difference of two infinite forgetting-costs; and the additive archetype $\infty - \infty$ already wears that shape. So in resistance coordinates the two archetypes are one [Fact, by hand], and the reading follows: **an indeterminate form is a contest between two infinite resistances that the value alone cannot settle.** What settles it is the *rate* at which each side approaches — and l'Hôpital's rule resolves the form by consulting exactly those rates, which is to say it reads the ledger. (Orders of infinity: du Bois-Reymond, Hardy.) [Reading]

The shadow point. The value is the shadow; the path is the process. An indeterminate form is a place where the shadow underdetermines the process — where the endpoint (0, 1, or ∞) leaves the outcome open and only the rate structure behind it decides. This is the same underdetermination the project reads at the barrier, now surfacing *inside* classical analysis rather than behind it. [Reading]

3. Quantization of hard erasure

The claim. A hard erasure event is the collapse of at least two distinguishable states onto one — fan-in ≥ 2 — so its cost is at least $\ln 2$, one genesis bit. Erasure is therefore quantized: in the genesis-bit machines below, any run deposits heat in integer multiples of $\ln 2$, and $\ln 2$ (fan-in 2) is the minimum quantum in general, since a fan-in- k erasure costs $\ln k \geq \ln 2$. [Fact]

The consequence. Because heat comes in whole genesis bits, a single-run heat reading H bounds the number of hard-erasure events on that run: at most $H / \ln 2$ of them. An infinite cascade of *actual* erasures cannot hide behind a finite reading — each event costs a full bit, and the bits add up. [Fact]

Verification. On the support of the cascade machine's exact distribution, every outcome's heat is an integer multiple of $\ln 2$ (support check: True). The point is conceptual; the check is the one line that secures it. [Fact]

The point (forward pointer). This is the wall the next sections press against. Actual erasure events are rigidly countable from a single run. What *can* hide — as §4 shows — is not an actual erasure but a *conditional* one, whose expected cost can be made arbitrarily small; and what exposes the hiding — as §5 shows — is not the mean but the variance. [Reading]

4. The series anchor and amortized hiding

The series anchor. [Fact] The genesis bit is itself an infinite sum of vanishing pieces: $\ln 2 = \sum_{n=1}^{\infty} 1/(n \cdot 2^n)$, verified to $|\text{diff}| = 0.000\text{e}+00$ over 200 terms. This is the Mercator series $-\ln(1-x) = \sum x^n/n$ at $x = 1/2$ — a convergent *resistance* series, a finite resistance assembled from unboundedly many terms. It is the hiding mechanism of §1 in its barest arithmetic form: finite magnitude over unbounded depth.

Two machines, one mean. [Fact] Put the mechanism to work.

- *Machine A* performs a single hard erasure of the genesis bit: heat = $\ln 2$ on every run, variance 0.
- *Machine B* runs an infinite cascade of *conditional* erasures: stage k fires — erasing one genesis bit — with probability 2^{-k} , independently. Its expected number of firings is $\sum 2^{-k} = 1$, so its expected heat is $\ln 2$ as well.

Their means coincide: $E[H_B] \rightarrow \ln 2 = E[H_A]$, converging to deviation 0.000e+00 by 64 stages. A probe that reads only average heat cannot tell the one hard erasure from the infinite cascade.

How the amortization works. [Fact] The equality of means is not because the machines behave alike. On Machine B a full $P(0 \text{ erasures}) \rightarrow \prod_{k=1} (1 - 2^{-k}) = 0.2888$ of runs fire *nothing at all* — over a quarter deposit zero heat — while rare runs fire several stages. The finite mean $\ln 2$ is amortised out of many do-nothing runs and a thin tail of do-a-lot runs. The infinite machinery is real on any given run; it is the *averaging* that folds it into a single finite number.

Reading. This is how a barrier hides depth from a mean-probe. The convergent resistance series is the mechanism, and the ensemble mean is the blind instrument: reading only the average, a probe sees a finite genesis bit and cannot distinguish a single erasure from an unbounded conditional cascade behind it. The nesting is not destroyed — it is amortised. [Reading]

5. The fluctuation probe

The mean was blind; the variance is not. [Fact] The two machines of §4 share a mean but not a variance.

- Machine A (one hard erasure) is deterministic: $\text{Var}[H_A] = 0$ exactly.
- Machine B (the infinite conditional cascade) fluctuates: $\text{Var}[H_B] \rightarrow (2/3) \cdot \ln^2 2 = 0.3203$, computed exactly by dynamic programming over the cascade's distribution — no sampling.

The first cumulant, the mean, cannot separate the machines — both are $\ln 2$. The second cumulant, the variance, separates them cleanly — 0 against $(2/3) \cdot \ln^2 2$. The probe that was blind at first order sees at second order.

Where the number comes from. [Fact] Heat is $m \cdot \ln 2$ for an integer firing-count m , so $\text{Var}[H] = \ln^2 2 \cdot \text{Var}[m]$. For independent stages with $P(\text{fire at } k) = 2^{-k}$, $\text{Var}[m] = \sum_{k=1} k^2 2^{-k} (1 - 2^{-k}) = \sum 2^{-k} - \sum 4^{-k} = 1 - 1/3 = 2/3$. Hence $\text{Var}[H_B] = (2/3) \cdot \ln^2 2$, matching the DP to the printed precision.

Reading — the fluctuation is the granularity showing through. A nonzero variance is the discreteness of §3 made visible. A single deterministic erasure has no spread; the cascade's spread is exactly the ensemble's uncertainty about *how many* integer bits fired. So the second cumulant does two things at once: it exposes the hidden nesting the mean amortised away, and it does so precisely because erasure is quantized — there is an integer count to be uncertain about. The mean sees the total; the variance sees that the total was assembled from discrete, countable events. [Reading]

Forward pointer (to §6). If the first cumulant is blind and the second sees, it is natural to ask whether each further cumulant is a further rung — whether matching d cumulants is what it costs

to hide depth d . That is the cumulant-ladder Lead, tested next. [Lead — stated, resolved in §6]

6. A parked Lead: the cumulant ladder

The Lead. §5 gave the first rung — the mean is blind, the variance sees. The ladder Lead generalises: for each depth d , do erasure machines exist that agree on cumulants $\dots_d$ and first differ at $_{d+1}$, so an order- d probe (one reading the first d cumulants) is fooled and the $(d+1)$ -th cumulant is the rung that exposes it — and does climbing to depth d carry a cost that grows with d ? Promotion was reserved for the case where the erasure-cascade structure adds something moment theory does not; anything explicable by classical moment theory parks. Expectation, disclosed in advance: park.

Closed-form cascade cumulants. [Fact] The genesis-bit cascade — independent stages k , stage k firing with probability 2^{-k} — has exact cumulants, from $_j = \Sigma_k \text{Bernoulli}(2^{-k})$ and $\Sigma_k 2^{-jk} = 1/(2^j - 1)$:

$$= 1, \quad = 2/3, \quad = 2/7, \quad = -2/105, \quad = -18/217, \quad = 58/651,$$

with denominators products of $(2^j - 1)$. The second cumulant recovers §5 exactly: $\cdot \ln^2 2 = (2/3) \cdot \ln^2 2$, the heat variance. This clean sequence is worth banking on its own.

The ladder exists. [Fact, classical] For $d = 1..4$ there are distribution pairs — built exactly in rationals by the finite-difference perturbation $w_i = (-1)^i C(d+1, i)$ on $\{0..d+1\}$, whose moments $0..d$ vanish and whose $(d+1)$ -th does not — that agree on $\dots_d$ and differ at $_{d+1}$. An order- d probe cannot separate them; the $(d+1)$ -th cumulant does. This is standard moment theory: cumulants and moments are interconvertible, and moment-matched distributions differing beyond a fixed order are classical.

The cost, in two currencies. [Fact] To hide depth d — to exhibit a distinct pair matching $\dots_d$ — costs:

currency	minimal size	source
free distribution (support points)	$d/2 + 1$	Gauss quadrature (s points fix $2s - 1$ moments)
genesis-bit cascade (stages)	$d + 1$	Newton's identities (a size- s multiset is fixed by its first s power sums)

The two coincide at $d = 1$ (both 2) and **diverge for $d \geq 2$** : the cascade costs $d + 1$ stages against the free distribution's $d/2 + 1$ support points. Hiding depth d in genuine independent stages is strictly more expensive than in a free distribution, because the cascade stages are rigid — a sum of independent Bernoullis (a Poisson binomial) carries fewer degrees of freedom than an arbitrary distribution on the same support.

Verdict — parked. The ladder is real, and the cascade cost is a clean, framework-native law: **depth d $d + 1$ genesis-bit stages**. But by the pre-registered criterion this parks rather than promotes. Both cost bounds are consequences of classical theory — Gauss quadrature and Newton's

identities — and the cascade-vs-free divergence, clean as it is, is itself classical: the rigidity of Poisson-binomial sums, not a law beyond moment theory. What the ladder shows is the framework’s arithmetic mapping cleanly onto the classical moment problem; it does not add to it. Newton’s identities in cascade dress. **[Lead — parked]**

Follow-up (F2): the heat-currency cost also parks. §6 priced hiding in *stages* and left the framework-native heat currency unmeasured. One route was closed in advance — expected heat is $\cdot \ln 2$, equal by construction, so it cannot be the price — leaving the **concealed leg** $H(\text{pattern} \mid \text{heat})$: how much of a hider’s internal firing pattern stays hidden behind its heat reading. (Patterns determine heat, so this is the chain rule, not a finding; the content is how it scales.) Across matched $(d+1)$ -stage machines for $d = 1..5$ ($\dots$ d matched, $\dots$ $\{d+1\}$ differing) and two canonical cascades computed exactly, the concealed leg $C = H(\text{pattern}) - H(\text{heat})$ is dominated by $H(\text{pattern}) = \sum_k h(q_k)$, while $H(\text{heat})$ grows only logarithmically. Normalising against the configuration count ($C / (s \cdot \ln 2)$) returns nothing but the stages’ own entropy budget: it climbs toward 1 for fair-coin stages ($q_k = \frac{1}{2}$) and *falls toward 0* for the genesis-bit cascade ($q_k = 2^{-k}$), whose deep stages fire too rarely to carry entropy; the concealed fraction $C / H(\text{pattern})$ tells the same story. **[Fact]** The heat-currency cost of hiding reduces to stage-entropy bookkeeping — not a law beyond the stage count — so §6’s park stands unimproved.

Two facts and a reading fall out. **Free-support machines have no concealed leg at all:** their outcome *is* the count *is* the heat, so $C = 0$; concealment is a cascade-specific phenomenon, and the currency matters. **[Fact]** Matching more cumulants **squeezes the hider toward the reference:** the total-variation distance between their heat distributions falls geometrically with d (0.020, 0.040, 0.008, 0.0016, 0.00032 across $d = 1..5$). **[Fact]** And, tying back to §§4–5: the genesis-bit cascade — Machine B itself — has a **bounded** concealed leg even at infinite depth. Its deep stages fire so rarely that its total *concealable* information saturates; Machine B hides its depth from the mean, but there is only a finite amount it can ever hide. **[Reading]** (Appendix F2.)

7. A parked Lead: the primordial indeterminate

The Lead. §2 read every indeterminate form as a contest between two infinite resistances that the value cannot settle. This Lead asks whether the genesis event itself — the first distinction emerging from non-distinction — carries such a form: does the pre-distinction state admit a formal residue in the pole-arithmetic frame?

The formal residue. **[Reading]** Under the resistance map $r = -\ln v$ the reciprocal poles 0 and ∞ sit at $+\infty$ and $-\infty$, and 1 — the unit, zero resistance — sits between them. Non-distinction is the absence of any resistance difference: the state in which the two infinite poles have not yet separated into a determinate relation. That is exactly the archetype §2 isolated — $0 \cdot \infty$, equivalently $\infty - \infty$ in resistance coordinates. So the structure of §2 extends, of its own accord, to a *form of the genesis event*: the first distinction is the resolution of a primordial $0 \cdot \infty$ into a determinate value, and the rate that resolves it — l’Hôpital reading the ledger — is that first distinction being recorded. The indeterminate form is the “before”; the recorded value is the “after”; the ledger entry is the resolution.

Why this is a residue and not a result. Two things keep this a Reading, held inside the axioms.

- *The pre-distinction state is behind the barrier.* By the probe axiom, the genesis layer lies

beyond any probe, and the pre-distinction state is the least probeable thing in the framework. To *assert* that it wears the form $0 \cdot \omega$ — that this is its structure, verifiable — would be a claim about what lies behind the barrier beyond the axioms that posit it, which the note's boundaries forbid. What we have is a structural extension of the shadow-side indeterminate-forms picture, offered as an analogy the frame supports, not a fact established about the layer.

- *An indeterminate form is, by definition, unresolved.* The primordial form's whole content is that it does not fix its own value; naming it does not resolve it, and could not, without supplying the very rate-structure — the ledger — whose arrival *is* genesis. The residue names the shape of the question, not an answer.

Verdict — parked, within the axioms. The formal residue is real as a structural reading: the pole-arithmetic frame of §2 reaches naturally to a form for the genesis event, $0 \cdot \omega$, resolved into a value by the first ledger entry. It is held as a **[Reading]** and parked — consistent with the framework's posit and claiming nothing about the layer beyond it. It is the frame's own account of where the indeterminate-forms story points when extended inward to genesis, not a probe of what cannot be probed. **[Lead — parked]**

8. Regimes, honest tensions, and what the note establishes

The regime distinction, as the note ran it. This note was the Regime-1/Regime-2 distinction at work. Forward from genesis (Regime 1): erasure events are discrete and integer, one genesis bit at a time — §3's quantization is that face, and it makes actual erasures rigidly countable from a single run. Backward from the probe (Regime 2): every heat *reading* is an ensemble average, and it is at this level that the hiding (§4) and its undoing (§5) live. The mean, a Regime-2 quantity, cannot see the Regime-1 granularity; the variance, also Regime-2, can — because the granularity is integer, there is a whole number to be uncertain about. The continuum is probe-manufactured; the system's own ledger is integer multiples of the genesis bit.

The tensions, collected.

- *Exact identities are instruments, not findings.* $\ln 2 = \sum 1/(n \cdot 2^n)$, $(2/3) \cdot \ln^2 2$, the eight indeterminate forms, the cumulant sequence — all exact by computation. The content is what they reveal — a mean cannot separate one erasure from an infinite cascade, a variance can, path-dependence picks out the classical indeterminates — never the identities themselves.
- *The machines are toy constructions.* Machines A and B are chosen to make a conceptual point exactly computable; no physical or thermodynamic measurement is asserted.
- *Precision.* A reported 0.000e+00 means double-precision agreement; the genuinely exact results are the ones checked in rationals or symbolically — the cumulants, the ladder, and the indeterminate-forms match.
- *§6 parked.* The cumulant ladder is classical moment theory — Newton's identities and Gauss quadrature; the cascade cost law is clean but adds nothing beyond them.
- *§7 parked, within the axioms.* The primordial $0 \cdot \omega$ is a structural reading the frame offers, not a claim about the pre-distinction state, which sits behind the barrier by the probe axiom.
- *Cartography, not evidence.* Everything is Regime-2 and shadow-side; the pole arithmetic and the erasure machines say nothing about whether the genesis system holds beyond the axioms that posit it.

What the note establishes. The note’s Facts are exact and modest. Erasure is quantized in genesis bits, so actual events are countable from a single run (§3); an infinite conditional cascade shares its mean with a single erasure (§4); the variance separates them, exactly $(2/3) \cdot \ln^2 2$ (§5); the cascade has closed-form cumulants (§6); and — the result that stands most on its own — path-dependence over the poles $\{0, 1, \infty\}$, defined with no prior notion of indeterminacy, reproduces exactly the eight classical indeterminate forms, which collapse under the resistance map to a single archetype: a contest between two infinite forgetting-costs (§2). The Readings that frame these — convergent resistance series as the barrier’s hiding mechanism, the cumulant ladder as the probe’s route in, the indeterminate form as shadow-underdetermination surfacing inside analysis, the primordial $0 \cdot \infty$ as the frame’s form for genesis — are held as Readings; the two Leads are parked; and no claim is made for or against the genesis system beyond the axioms that posit it. The same discipline, and the same limits, as the rest of the project.

Citations.

- *Thermodynamics of computation.* The erasure cost is Landauer (1961); the cost-free status of copying is Bennett (1973); the fluctuation relation connecting nonequilibrium work to free-energy differences is Jarzynski (1997), the project’s grounding for reading fluctuations physically.
- *Information theory and the moment problem.* Entropy and conditional entropy are Shannon (1948); cumulants, the moment problem, Gauss quadrature, and Newton’s identities on power sums and elementary symmetric functions are classical.
- *Orders of infinity.* The rates that resolve indeterminate forms trace to du Bois-Reymond and to Hardy, *Orders of Infinity* (1910); l’Hôpital’s rule is classical.
- *The convergent-series emblem.* Chaitin’s Ω (the halting probability) is the emblem of finite magnitude over unbounded depth; $\ln 2 = \sum 1/(n \cdot 2^n)$ is the Mercator series at $x = 1/2$.
- *Framework.* The resistance isomorphism $\Omega(1/n) = \ln n$, the genesis bit $\ln 2$, the Regime-1/Regime-2 distinction, and the two-legged ledger are developed in the project’s Papers 1–9 and companion notes, including the companion “Proofs on the Ledger.”

Appendix — verbatim script outputs

Built incrementally, one block per experimental step; every number in the prose above appears here.

B2 · indeterminate_poles.py

a	op	b	limits along paths	
0	-	0	determinate	['0']
0	-	1	determinate	['-1']
0	-	inf	determinate	['-oo']
0	*	0	determinate	['0']
0	*	1	determinate	['0']
0	*	inf	INDETERMINATE	['0', '1', 'oo']
0	/	0	INDETERMINATE	['0', '1', 'oo']
0	/	1	determinate	['0']
0	/	inf	determinate	['0']
0	^	0	INDETERMINATE	['1', 'exp(-1)']


```

0 ^      1  determinate  ['0']
0 ^    inf  determinate  ['0']
1 -      0  determinate  ['1']
1 -      1  determinate  ['0']
1 -    inf  determinate  ['-oo']
1 *      0  determinate  ['0']
1 *      1  determinate  ['1']
1 *    inf  determinate  ['oo']
1 /      0  determinate  ['oo']
1 /      1  determinate  ['1']
1 /    inf  determinate  ['0']
1 ^      0  determinate  ['1']
1 ^      1  determinate  ['1']
1 ^    inf  INDETERMINATE ['0', 'E', 'exp(-1)', 'oo']
inf -      0  determinate  ['oo']
inf -      1  determinate  ['oo']
inf -    inf  INDETERMINATE ['-oo', '0', 'oo']
inf *      0  INDETERMINATE ['0', '1', 'oo']
inf *      1  determinate  ['oo']
inf *    inf  determinate  ['oo']
inf /      0  determinate  ['oo']
inf /      1  determinate  ['oo']
inf /    inf  INDETERMINATE ['0', '1', 'oo']
inf ^      0  INDETERMINATE ['1', 'E']
inf ^      1  determinate  ['oo']
inf ^    inf  determinate  ['oo']

```

Path-dependent set found: [('0', '*', 'inf'), ('0', '/', '0'), ('0', '^', '0'), ('1', '^', 'in
Classical indeterminate: [('0', '*', 'inf'), ('0', '/', '0'), ('0', '^', '0'), ('1', '^', 'in
Exact match: True

B3-B5 · heat_hiding_fluctuations.py (single run; sliced across §§3-5)

This one run supplies the quantization support check (§3), the series anchor and cascade means (§4), and the fluctuation/variance results (§5).

(1) $\sum 1/(n \cdot 2^n) = 0.693147180559945$; $\ln 2 = 0.693147180559945$; $|\text{diff}| = 0.000e+00$

(2)+(4) Machine B cascade, exact (heat = $m \cdot \ln 2$ per run):

stages	K	E[H]	E[H]-ln2	Var[H]	P(0 erasures)
	2	0.519860385	-1.733e-01	0.210198194	0.375000000
	4	0.649825482	-4.332e-02	0.290899286	0.307617188
	8	0.690439574	-2.708e-03	0.318427683	0.289919118
	16	0.693136604	-1.058e-05	0.320294678	0.288792502
	32	0.693147180	-1.614e-10	0.320302009	0.288788095
	64	0.693147181	0.000e+00	0.320302009	0.288788095

Predicted Var limit = $\ln 2^2 \cdot (1 - 1/3) = 0.320302009$ (= $(2/3) \ln 2^2 = 0.320302009$)

Machine A: $E[H] = 0.693147181$, $\text{Var}[H] = 0$ exactly

- (3) Every run's heat is an integer multiple of $\ln 2$ (support check): True
 $\Rightarrow$ a single-run heat reading H bounds the hard-erasure count by $\text{floor}(H/\ln 2)$;
only the ENSEMBLE MEAN is blind to the nesting; the variance is not.

B6 · cumulant ladder (self-authored, PID-named; parked Lead)

cross-check ($\cdot \ln^2 2 = 0.320302009$) reproduces §5's heat variance, tying the cumulant computation to the fluctuation result.

- (1) EXACT cascade cumulants (κ_j , from $\text{sum}_k \text{Bernoulli}(2^{-k})$):
 $\kappa_1 = 1 = +1.000000000$
 $\kappa_2 = 2/3 = +0.666666667$
 $\kappa_3 = 2/7 = +0.285714286$
 $\kappa_4 = -2/105 = -0.019047619$
 $\kappa_5 = -18/217 = -0.082949309$
 $\kappa_6 = 58/651 = +0.089093702$
cross-check: $\kappa_1=1.000000 (=1)$, $\kappa_2 \cdot \ln^2 2 = 0.320302009$ (heat variance $(2/3)\ln^2 2$)
- (2) Ladder exists (EXACT, general distributions on $\{0..d+1\}$):
 $d=1$: $\kappa_1.. \kappa_1$ match: True; κ_2 differs: True ($kP[2]= 3/4$, $kQ[2]= 1/4$)
 $d=2$: $\kappa_1.. \kappa_2$ match: True; κ_3 differs: True ($kP[3]= -3/8$, $kQ[3]= 3/8$)
 $d=3$: $\kappa_1.. \kappa_3$ match: True; κ_4 differs: True ($kP[4]= 1/4$, $kQ[4]= -5/4$)
 $d=4$: $\kappa_1.. \kappa_4$ match: True; κ_5 differs: True ($kP[5]= -15/8$, $kQ[5]= 15/8$)
- (3) Cost of hiding depth d : free-support bound vs cascade-stage bound

d	$s_{\text{free}}=\text{ceil}(d/2)+1$	$s_{\text{cascade}}=d+1$	distinct	$(d+1)$ -stage pair matches $k1..kd$?
1	2	2	True	(d -stage pair exists: False)
2	2	3	True	(d -stage pair exists: False)
3	3	4	True	(d -stage pair exists: False)
4	3	5	True	(d -stage pair exists: False)

minimality of $s_{\text{cascade}}=d+1$ is Newton's identities (size- s multiset fixed by $P_1..P_s$);
 $s_{\text{free}}=\text{ceil}(d/2)+1$ is Gauss quadrature (s points fix $2s-1$ moments). Both are classical.

F2 · concealed-leg / heat-currency cost (self-authored, PID-named; parked, refines §6)

Concealed leg $C = H(\text{pattern}) - H(\text{heat})$, in nats. Distributions exact-rational where probabilities are rational (Part 1); matched pairs numerical (Part 2). Normalised concealed leg reduces to stage-entropy bookkeeping; free-support machines have $C = 0$; the hider-reference TV distance falls with d .

PART 1 Canonical cascades, exact rational distributions

family	s	$H(\text{pattern})$	$H(\text{heat})$	C	$C/(s \ln 2)$	$\phi=C/H_{\text{pat}}$
genesis 2^{-k}	2	1.25548	0.97431	0.28117	0.20282	0.22395
genesis 2^{-k}	3	1.63225	1.08442	0.54783	0.26345	0.33563

genesis 2^{-k}	4	1.86604	1.13286	0.73318	0.26444	0.39291
genesis 2^{-k}	5	2.00510	1.15584	0.84926	0.24505	0.42355
genesis 2^{-k}	6	2.08559	1.16707	0.91852	0.22086	0.44041
fair-coin $1/2$	2	1.38629	1.03972	0.34657	0.25000	0.25000
fair-coin $1/2$	3	2.07944	1.25548	0.82396	0.39624	0.39624
fair-coin $1/2$	4	2.77259	1.40753	1.36506	0.49234	0.49234
fair-coin $1/2$	5	3.46574	1.52367	1.94207	0.56036	0.56036
fair-coin $1/2$	6	4.15888	1.61736	2.54152	0.61111	0.61111

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PART 2 B6 matched pairs: (d+1)-stage reference vs hider matching kappa_1..kappa_d
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d	stages	kmatch	kdifff	C_ref	C_hid	Cref/(s ln2)	phi_ref	TV(ref,hid)
1	2	True	True	0.1521	0.1699	0.1097	0.1520	0.020000
2	3	True	True	0.5611	0.5633	0.2698	0.3313	0.040000
3	4	True	True	1.0433	1.0501	0.3763	0.4446	0.008000
4	5	True	True	1.5602	1.5602	0.4502	0.5220	0.001600
5	6	True	True	2.0985	2.0986	0.5046	0.5787	0.000320

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PART 3 Free-support hider contrast
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d=1: free-support machine needs $\text{ceil}(d/2)+1 = 2$ points; outcome = count = heat -> H(patter
d=2: free-support machine needs $\text{ceil}(d/2)+1 = 2$ points; outcome = count = heat -> H(patter
d=3: free-support machine needs $\text{ceil}(d/2)+1 = 3$ points; outcome = count = heat -> H(patter
d=4: free-support machine needs $\text{ceil}(d/2)+1 = 3$ points; outcome = count = heat -> H(patter
d=5: free-support machine needs $\text{ceil}(d/2)+1 = 4$ points; outcome = count = heat -> H(patter

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READINGS
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C is dominated by $H(\text{pattern}) = \sum h(q_k)$; $H(\text{heat}) \leq \log(s+1)$ grows only logarithmically.
 $C/(s \ln 2) \rightarrow$ the average stage entropy / $\ln 2$ (fair-coin $\rightarrow \sim 1$; genesis $\rightarrow \rightarrow 0$ as it saturates
 $\Rightarrow$ normalized concealed leg reduces to stage-count bookkeeping. Free-support $C=0$.

End of Note B (draft). Sections 1–8 complete, with follow-up test F2 (appendix) refining the §6 park in the heat currency. The appendix carries every number quoted in the prose. Two Leads parked — §6 (F2-refined) and §7. Nothing here is promoted to the project spine.